

Dr Samuel Ngigi Mbote, MBChB, MCS

Molecular & Structural Biology Expert | Computational Biochemistry | AI Scientific-Reasoning Evaluator

mbotesamuel9@gmail.com | linkedin.com/in/samuel-mbote-238b13230 | github.com/mbote-droid

Professional Summary

Physician-scientist with deep, applied expertise in **molecular and structural biology** and a rare, directly-relevant edge: I have spent 2026 **building and auditing AI systems that reason over biochemistry**. My work spans **protein folding and structure** (AlphaFold, the ESM-2 protein language model, per-residue pLDDT confidence, $\Delta\Delta G$ stability), **drug-target interactions and enzyme inhibition** (molecular docking, MDM2 inhibitors, synthetic lethality), and **molecular genetics and biochemical pathways** (variant effect, cell-cycle, apoptosis, DNA-repair). Crucially, I already do this role's core function — I built an **AI evaluation layer that audits AI-generated biochemical reasoning** for technical accuracy, factual concordance, and logical soundness. IBM-certified in Python for Data Science & AI; strong scientific-writing and peer-review discipline across 480+ tested, documented deliverables.

Direct Fit for This Role

- **Auditing AI biochemical reasoning** — built a dual-gate evaluation engine and a per-agent harness that flag hallucinations, factual errors, and unsupported claims in AI-generated molecular analyses — the exact "audit AI outputs / correct reasoning errors" task.
- **Structural biology & protein folding** — AlphaFold-predicted structures with pLDDT confidence, ESM-2 masked-marginal variant-effect scoring, $\Delta\Delta G$ destabilisation, domain and drug-binding-site analysis.
- **Drug-target & enzymology** — molecular docking (AutoDock Vina), MDM2/p53 inhibitor scoring (APR-246/PRIMA-1), PARP synthetic-lethality mapping, cavity druggability.
- **Molecular genetics & pathways** — variant biochemistry (HGVS, hotspot classification), cell-cycle / apoptosis / DNA-repair pathway perturbation, cross-species conservation.
- **Precise technical writing & ground-truth authoring** — audit-grade documentation, benchmark datasets vs ClinVar/IARC, structured step-by-step scientific reasoning.

Signature Computational Biochemistry & AI Projects

Precision Onco Africa — Lead Architect & AI Reasoning Evaluator (2026)

Structural & molecular-genetics AI platform | 419 automated tests | github.com/mbote-droid/precision-onco-africa

- Built **protein structure & folding analysis** around AlphaFold-predicted p53 structures (per-residue pLDDT), the **ESM-2 protein language model** for masked-marginal variant-effect scoring, and $\Delta\Delta G$ structural-stability estimation with cavity/druggability analysis.
- Modelled **drug-target interactions** — molecular docking, MDM2 inhibitor and APR-246 candidate scoring, and PARP-inhibitor **synthetic-lethality** mapping grounded in trial evidence.
- Engineered the **AI-audit layer this role is about**: a dual guardrail (syntactic + factual ClinVar cross-check) plus an evaluation harness scoring every AI agent on **accuracy, hallucination rate, and citation quality**, catching reasoning errors before release.
- Authored **ground-truth benchmarks vs ClinVar/IARC (100% pathogenic-variant F1)**; mapped variant effects onto **cell-cycle, apoptosis, and DNA-repair pathways** with explicit uncertainty.
- 26-agent system, HIPAA-aligned, HL7 FHIR R4, deployed on AMD Instinct GPUs; runs fully offline on 8 GB RAM.

TP53 Bioinformatics Analysis Pipeline — Lead Developer (2026)

Live deployment | 63 tests | CI/CD | tp53analysis.streamlit.app

- Nucleic-acid and protein workflows: NCBI sequence retrieval, DNA translation, **6-frame ORF discovery**, pairwise alignment, **multi-species phylogenetics** (Neighbor-Joining across human, mouse, rat, chimp, zebrafish), and EMBL-EBI **InterPro domain annotation** (25 confirmed TP53 domain hits).
- Cancer-mutation biochemistry heatmap across 12 hotspot residues $\times$ 11 cancer types; 63 unit tests + CI/CD.

Healthcare AI Radar — Creator & Lead Engineer (2026)

Scientific-literature evaluation pipeline | 58+ tests | github.com/mbote-droid/healthcare-ai-radar

- Automated ingestion and **quality-evaluation** of scientific literature (Nature Medicine, FDA, arXiv, PubMed) with a **five-layer anti-hallucination LLM guardrail** — live-verified to produce zero fabricated content.

Domain Experience

Physician-Scientist & General Surgery Resident (COSECSA-Accredited) | 2022–Present

Embu Teaching and Referral Hospital, Embu, Kenya

- **7+ years** of applied clinical and molecular-oncology reasoning across colorectal, hepatobiliary, and breast malignancies — deep, verifiable command of molecular pathology and the biochemistry of disease.
- Rigorous, detail-oriented scientific documentation with **100% accuracy across hundreds of cases**; sharp eye for logical and factual error — the core competency of AI reasoning-evaluation.
- **COSECSA Member (Part 1, Dec 2025)** — credentialed scientist within an internationally recognised surgical college.

Education & Certifications

MBChB (Bachelor of Medicine & Surgery) — Kampala International University | **MCOSECSA** — College of Surgeons of East, Central & Southern Africa (2025) | Whole Genome Sequencing (Certified) | IBM Python for Data Science, AI & Development | IBM Cloud Computing & DevOps | Image/Object Annotation, Detection & Segmentation | Google Prompting Essentials | Git & GitHub. **In progress:** IBM AI Engineering, Microsoft Azure AI in Healthcare, Microsoft AI Agents.

Technical Skills

Structural & Computational Biochemistry: protein folding & structure (AlphaFold, ESM-2 protein language model, pLDDT, $\Delta\Delta G$) | molecular docking & drug–target interaction (AutoDock Vina) | enzyme-inhibitor & synthetic-lethality analysis | domain annotation (InterPro) | AlphaFold/BLAST/PyMOL-context

Molecular Genetics & Bioinformatics: variant biochemistry (HGVS, ClinVar/IARC) | pathway analysis (cell-cycle, apoptosis, DNA repair) | BioPython | NCBI Entrez | 6-frame ORF, pairwise alignment, phylogenetics (NJ/IQ-TREE) | codon-usage bias

AI Reasoning Evaluation: AI output auditing for scientific soundness | hallucination & error detection | ground-truth benchmark authoring | dual-guardrail validation | data annotation & quality evaluation | prompt-injection testing

Scientific Communication: precise technical writing | audit-grade documentation | benchmark/dataset authoring | peer-review-grade rigor

Languages & Tools: Python | SQL | Bash | Docker | GitHub Actions CI/CD | pytest (480+ tests across projects)

Availability

Self-directed and remote-native — **available 10–40 hrs/week, asynchronous, anywhere.** Comfortable authoring rigorous, independent scientific deliverables on my own schedule.